# Leaf Herbivory Lab Activity

## Physical Tasks

Take the 20 leaves you and your partner collected and use the Leafbyte App (free on IOS) to estimate the leaf area (cm2) and the percent consumed (%). Use a balance to determine the mass of the leaf (g).

## Digital Tasks

Input your data into the class spreadsheet! Once all groups have input their data, you will use that data to complete the rest of the assignment.

For this lab, we primarily want to think about whether leaf species determine the degree of herbivory.

- State an appropriate null and alternative hypothesis relating to this. (1 point)

Download the class data as a CSV file, and open it in JASP.

Let's test our hypotheses right off the bat!

- Complete the appropriate t-test within JASP. What was the p-value it gave you? We are going to stick with convention and use a significance level of p=0.05. Based on this, can we reject our null hypothesis? (1 point)

Usually before we jump into a t-test, we want to make sure the data meets the assumptions of the test.

- Select the two tests in JASP that are appropriate for testing our assumptions of normality and equal variance. What are the results of those tests? (1 point)

If our data is not normal, we can try to transform it, or we can use a different type of test - Which one? Whether our data is normal or not, go ahead and carry out this test and describe the results. (1 point)

This test tests the null hypothesis that the distributions are the same, and the alternative hypothesis is that the distributions are not the same. Therefore, even if we reject the null, this doesn't necessarily imply anything about the central tendency of the data.

- To take a closer look, generate a histogram of the percent herbivory for each species (% herbivory on the x axis, count on the y axis). Briefly describe/compare the histograms. (2 points)

Whether or not our data produced results that are statistically significant, we may still be able to see a trend in the data.

- Generate a boxplot of the percent herbivory by species and briefly describe it. Additionally, provide the mean and standard deviation of the percent herbivory by species. (2 points)

After our brief investigation, does leaf species appear to affect the pattern of herbivory in this case? Can you look up the characteristics of these plant species (or take into account your observations of the plants you sampled) and hypothesize one or two traits about either that might suggest an explanation for the pattern you observe? (2 points)

Last thing: Pick one of the other aspects of the leaves that we measured, and repeat these tests on it as an informal investigation. No need to provide any graphs, but do give a brief explanation of your results (3-4 sentences). (2 points)

## Deliverables

- A PDF of your JASP results that includes your t-test, assumptions, histograms, boxplot, and summary statistics.
- Answers to all of the questions above, in a PDF/word document/other format

It is up to you whether you decide to include your JASP file or not, but it can be helpful in case your results PDF leaves something out accidentally (the JASP file can be downloaded and checked).

This is due on Thursday, 09/17 at 11:59 PM.